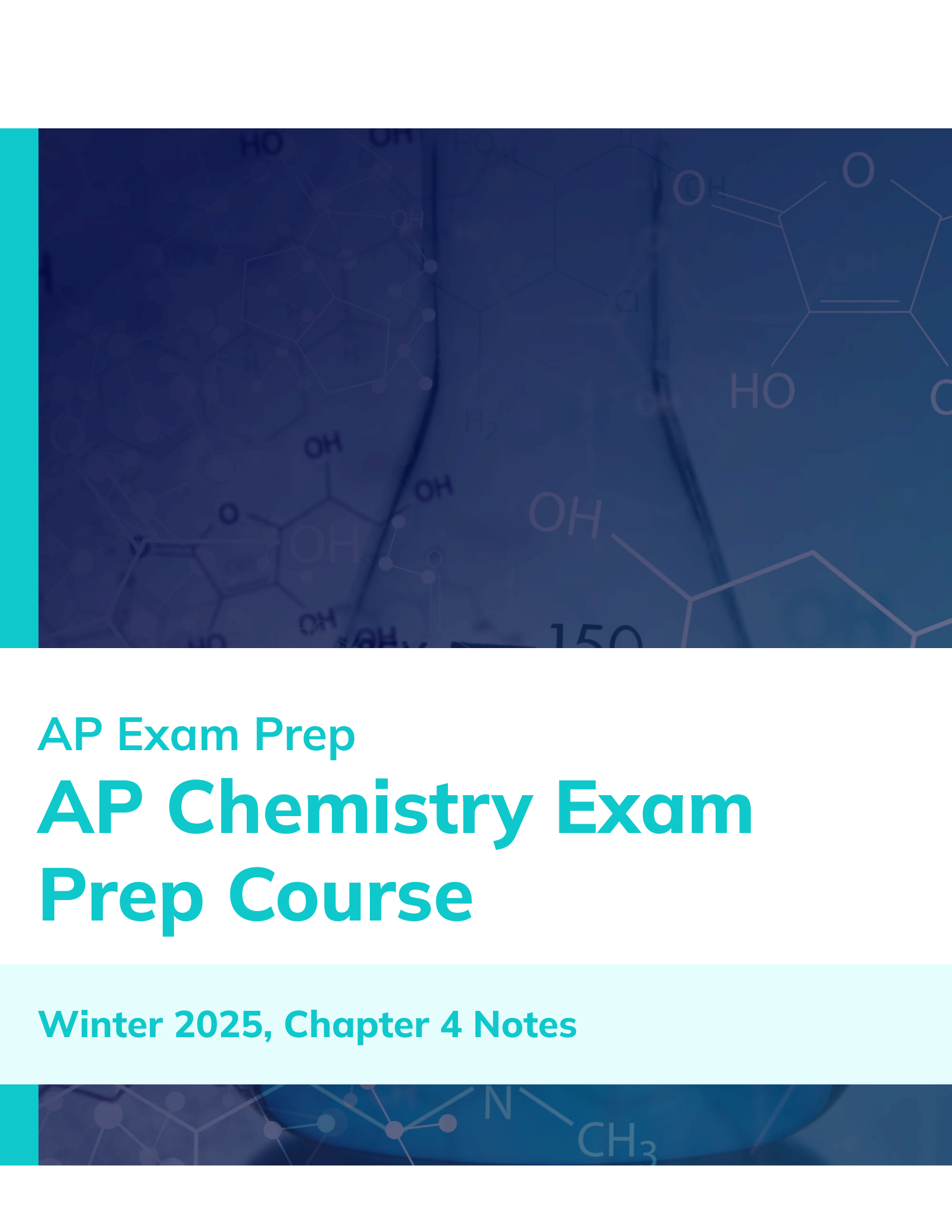




AP Exam Prep

AP Chemistry Exam Prep Course

Winter 2025, Chapter 4 Notes

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4. Chemical Reactions

4.1

Reactions & Energy Changes

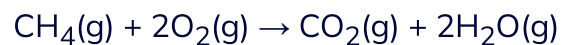
4.1 Introduction for Reactions

4.1.1

Chemical vs Physical Changes

A **chemical change** is when bonds are broken and formed between different atoms

Example: Burning a compound in a chemical reaction is an example of a chemical change. The compound is changed as a result of the burning and can't be changed back (we **get a new compound!**)



A **physical change** does not involve the breaking and forming of bonds. Instead, these involve a **change of state**



Example: When ice melts, we have liquid water. Ice is $\text{H}_2\text{O}(\text{s})$ while water is $\text{H}_2\text{O}(\text{l})$. The compound is still the same and could be frozen again to become ice again!



4.1.2

Are the following changes physical or chemical changes?

A Change to a Compound	Chemical or Physical Change (Enter answer as C or P)
Seeing bubbles during a reaction	
Feeling heat on the outside of a beaker during a reaction	
Boiling the compound	
Seeing a precipitate form in solution	
Seeing light given off of the compound during the reaction	
Melting the compound	
Seeing a new compound form during the reaction	

4.2 Introduction to Chemical Equations and Balancing

4.2.1

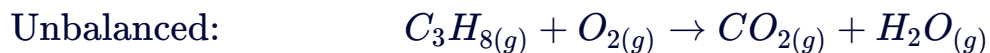
Balancing Reactions

Reactions must be balanced before the stoichiometry can be analyzed.

What makes a reaction balanced?

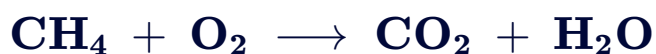
The number of _____ of each type on the _____ must equal the number of atoms of each type on the _____ of the reaction.

Example:



Balancing Chemical Reactions

Unbalanced Reaction



The **subscripts** tell us the ratio of different atoms in one molecule.

Example: One CH_4 molecule has 4 hydrogen atoms and 1 carbon atom

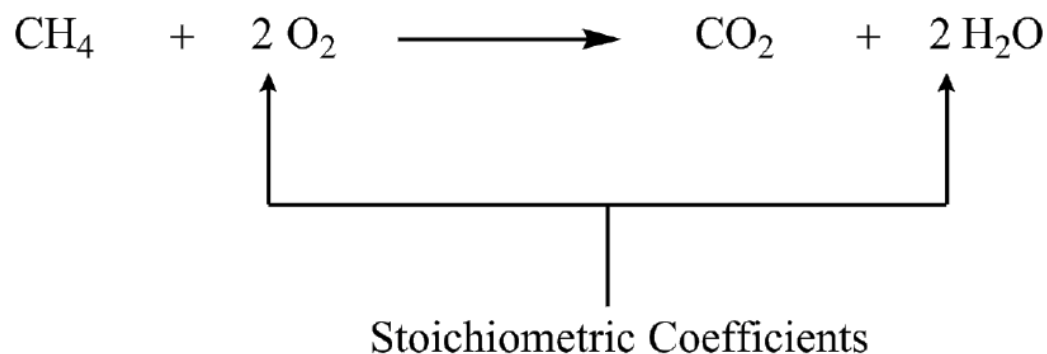
WATCH OUT!

Sometimes you could be given unbalanced equations and you are expected to balance them!
We **need to use the balanced equation for calculations** in order to get the correct answer!

Steps to Balance Chemical Equations

1. Balance **elements that only appear once** on the reactant side and once on the product side (typically elements not including C, N, H, O)
2. **Group polyatomic ions** and balance them as a group (not as separate elements, e.g. balance NO_3 as 1 NO_3 group, not 1 N and 3 O)
3. **Balance all other elements**, starting with the least common elements to the most common elements
4. Make sure all **coefficients are whole numbers** AND are represented in the **simplest integer ratio possible**
5. **Check** to make sure all elements are balanced on each side of the chemical equation

Balanced Reaction



- The **stoichiometric coefficients** of the balanced equation tell us that we need ____ oxygen molecules for each methane (CH_4) molecule and we produce ____ waters for every methane consumed.
- It also tells us that for every 1 mole of methane used we produce ____ mole of CO_2 .

WIZE CONCEPT

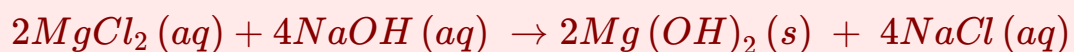
Multiply coefficients by subscripts to see exactly how much of an atom we have on each side.

! WATCH OUT!

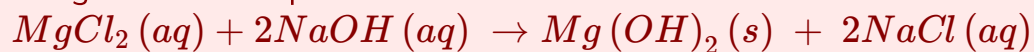
We want the **lowest possible coefficients**.

Example:

If I balanced an equation and got:



The **greatest common factor** that can go into each coefficient is 2 so we need to divide each coefficient by 2 to get the lowest possible numbers!



Example: Balancing a Chemical Reaction

Balance the following equation:



WIZE CONCEPT

The physical state of each chemical compound is often included in chemical equations:

(s) = solid

(l) = liquid

(g) = gas

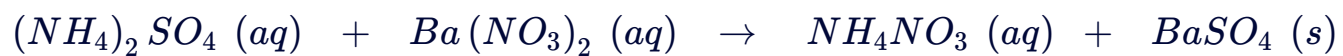
(aq) = aqueous

Polyatomic Ions

Note: There is no need to memorize the following table. It is just included here so you have an idea about what a polyatomic ion is!

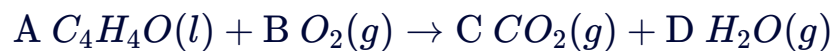
Polyatomic Ion Cheatsheet			
Ion	Name	Ion	Name
NH_4^+	Ammonium	MnO_4^-	Permanganate
NO_2^-	Nitrite	CrO_4^{2-}	Chromate
NO_3^-	Nitrate	$\text{Cr}_2\text{O}_7^{2-}$	Dichromate
SO_3^{2-}	Sulfite	O_2^{2-}	Peroxide
SO_4^{2-}	Sulfate	O_2^-	Superoxide
HSO_4^-	Hydrogen sulfate*	$\text{C}_2\text{O}_4^{2-}$	Oxalate
HSO_3^-	Hydrogen sulfite	HC_2O_4^-	Hydrogen Oxalate
OH^-	Hydroxide	HCO_2^-	Formate
CN^-	Cyanide	$\text{S}_2\text{O}_3^{2-}$	Thiosulfate
PO_4^{3-}	Phosphate	HS_2O_3^-	Hydrogen Thiosulfate
HPO_4^{2-}	Hydrogen Phosphate	BrO_4^-	Perbromate
H_2PO_4^-	Dihydrogen Phosphate	BrO_3^-	Bromate
CO_3^{2-}	Carbonate	BrO_2^-	Bromite
HCO_3^-	Hydrogen Carbonate**	BrO^-	Hypobromite
ClO^-	Hypochlorite	IO_4^-	Periodate
ClO_2^-	Chlorite	IO_3^-	Iodate
ClO_3^-	Chlorate	IO_2^-	Iodite
ClO_4^-	Perchlorate	IO^-	Hypoiodate
$\text{C}_2\text{H}_3\text{O}_2^-$	Acetate		

Example: Balance a Chemical Equation with Polyatomic Ions



Practice: Balancing the Combustion of Furan

Choose the answer that balances the reaction shown below:



A = 2; B = 9 ;C = 8; D = 4

☐

A = 1; B = 3 ;C = 5; D = 2

☐

A = 1; B = 1 ;C = 1; D = 1

☐

A = 4; B = 25 ;C = 20; D = 10

☐

4.3 Stoichiometry

4.3.1

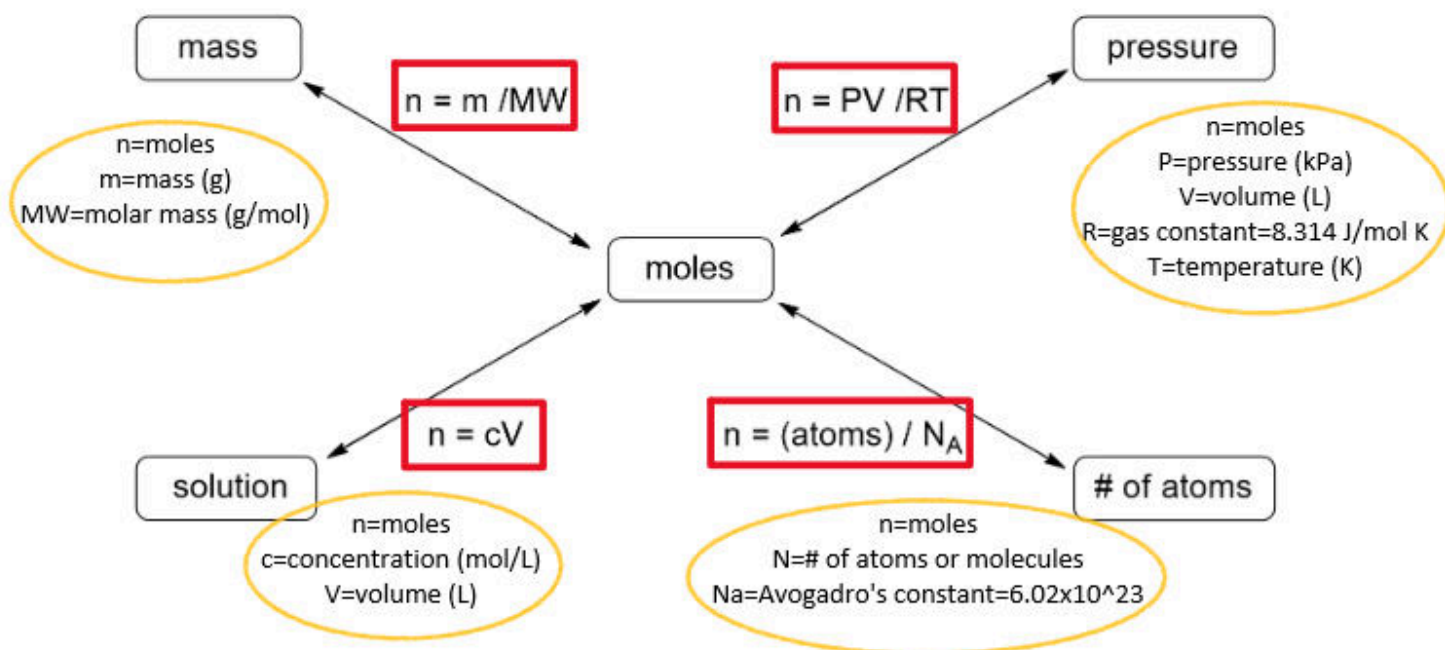
Intro to Stoichiometry

Stoichiometry allows us to predict the quantities of products or reagents across a chemical reaction.

Typically this is done in a few steps:

1. Calculating the number of moles of the reagents
2. Calculating the number of moles of the products using the stoichiometric ratio
3. Calculating the mass or pressure of the products.

Moles are the central unit!



In addition we can convert between mass and volume of a pure substance by looking at it's **density**,

$$\delta = m/V$$

Stoichiometry of a Reaction

We use the **coefficients** of the **balanced reaction** along with the our equations that convert mass, volume, and concentration into moles to predict the quantities of reactants and products in a chemical reaction.

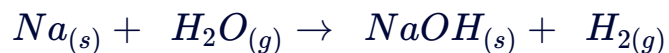
To answer any stoichiometry problem, focus on converting to and from moles!

General Steps to Solving a Stoichiometry Problem:

1. Convert the values given in the problem about a reactant or product to a **number of moles**
2. Use the **stoichiometric coefficients** from the **balanced** reaction to **find the number of moles of the unknown** you are being asked for
3. Convert the number of moles of your unknown to a mass, or whatever quantity you are being asked for

Example

2.6g of sodium metal (Na) reacts with water to form NaOH and H₂ according to the unbalanced reaction below. Once the reaction is complete how many grams of NaOH are formed?



WIZE TIP

On an exam, your prof will NOT specify if an equation is balanced or unbalanced.

Always double check that the equation is balanced. If it's not, balance the equation before continuing!

The equation must be balanced in order to get the correct answer!

Step 1: Balance the reaction



Step 2: Find the number of moles of sodium

$$n_{\text{Na}} = \frac{m}{MW} = \frac{2.6 \text{ g}}{22.990 \text{ g/mol}} = 0.113 \text{ mol}$$

Step 3: Find the number of moles of NaOH

$$n_{\text{NaOH}} = n_{\text{Na}} \times \frac{\text{coefficient NaOH}}{\text{coefficient Na}} = 0.113 \text{ mol} \times \frac{2}{2} = 0.113 \text{ mol}$$

Step 4: Convert the number of moles of NaOH to a mass

$$n = \frac{m}{M}$$

$$m_{\text{NaOH}} = n \times MW = 0.113 \text{ mol} \times 39.997 \text{ g/mol} = 4.520 \text{ g} = 4.5 \text{ g}$$

Example: Converting Mass of Reactant to Mass of Product

Calculate the mass of hydrogen gas produced if 0.550 g of iron powder is reacted with an excess amount of sulfuric acid.



Example: Solution Stoichiometry to Determine Mass of Product

Lithium metal was added to a 25mL of a 1.3M solution of Ag_2SO_4 . The unbalanced chemical reaction is shown below.

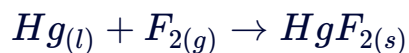


Once the reaction has gone to completion, what mass of silver metal is produced?

- a) 0.0070 g
- b) 7.0 g
- c) 3.5 g
- d) 0.0035 g

Practice: Calculate the Mass of Product

16.0 mL of Hg (mp = -38.8°C , density = 13.56 g/mL) reacts with fluorine gas and the reaction goes to completion as shown below.



Once the reaction has gone to completion, what mass of mercury fluoride, HgF_2 , is produced?

A) 1.48 g

☐

B) 514 g

☐

C) 142 g

☐

D) 258 g

☐

Practice: Stoichiometry Calculations Application

A solution that is 183mM (millimolar) NaCl(aq) is isoosmotic with plasma. This means that cells don't swell or shrink when in this solution. How many grams of sodium chloride are required to make 150mL of isoosmotic NaCl(aq)? The molar mass of NaCl is 58.44g/mol.

0.2g

☐

1.6g

☐

2.2g

☐

4.9g

☐

4.4 Limiting Reagents

4.4.1

Introduction to Limiting Reagents

Anytime **reactant species are in limited supply** and not present in perfectly proportional amounts, a chemical reaction will have a **limiting reagent** which will be totally **consumed before any other reactant**

The quantity of the limiting reagent available directly **determines the maximum number of product molecules** that can be formed!

Let's Consider an Example:

When making smores (yum!!) the "reaction" looks something like:



If I had 10 graham crackers, 6 pieces of chocolate, and 6 marshmallows, what would be the limiting reagent? In other words, what would I run out of first? And how many smores could I make?

- The "chemistry way" to figure this out would be to **take the # of moles of each reactant and divide by its stoichiometric coefficient**:

Graham crackers: $10/2=5$

Pieces of chocolate: $6/1=6$

Marshmallows: $6/1=6$

- Now, **to figure out the limiting reagent, look at which of the above numbers are the smallest!!**
 - 5 is smallest, therefore Graham crackers are the limiting reagent!!

Now how many smores could we make?

We know graham crackers will determine how much product we get since we'll run out of the crackers first.

$$\text{moles of graham cracker} \times \frac{1 \text{ mol smores}}{2 \text{ moles graham cracker}} = \text{moles of smores}$$

$10 \times (1/2) = 5$ moles of smores created!

 WIZE CONCEPT

It is necessary to **determine the limiting reagent** whenever we are **given the amounts of two or more reactants** in a chemical reaction.

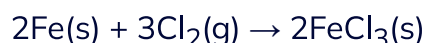
The limiting reagent is **totally consumed in the reaction**. **Any other reactant is in excess**.

Limiting reagent stops the reaction by **running out first**.

The **quantity of the limiting reagent** available directly **determines the maximum number of products** molecules that can be formed

Example: Determine the Mass of Product in a Limiting Reagent Problem

Iron and chlorine gas react to form iron (III) trichloride. If 110 g of iron and 105 g of chlorine gas are reacted, which species is the limiting reagent? What is the maximum mass of FeCl_3 that can be formed?



WIZE TIP

Steps for Solving Limiting Reagent Problems:

Step 1 – Write & **balance** the equation.

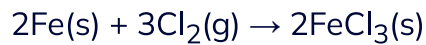
Step 2 – Calculate **moles of each reactant**.

Step 3 – Use the molar ratios of the present reactants (from the balanced equation) to determine the limiting reactant (LR).

- Take the **# of moles of each reactant** from step 2 and **divide by the stoichiometric coefficient** for that reactant
 - **Smallest value** from this calculation tells us the **LR**

Step 4 – From the limiting reactant, use the **molar ratio** (from the balanced equation) to **calculate moles of the desired product**.

Step 5 – Convert moles to the desired units (density, molarity, grams, etc...)



Step 1 – Write & balance the equation.

Is the equation given balanced? _____

Step 2 – Calculate moles of each reactant.

i) Find moles of Fe

ii) Find moles of Cl_2

Step 3 – Use the molar ratios of the present reactants (from the balanced equation) to determine the limiting reactant.

Therefore the limiting reagent is _____ ! (because it had the smallest number)

Step 4 – From the limiting reactant, use the molar ratio (from the balanced equation) to calculate moles of the desired product.

- The maximum mass of FeCl_3 that is formed depends on how much _____ we have (LR).

Step 5 – Convert moles to the desired units (density, molarity, grams, etc...)

Example: What is the Limiting Reagent?

2.0 g of aqueous Barium hydroxide ($\text{Ba}(\text{OH})_2(\text{aq})$, 171.32 g/mol) and 1.5 g of liquid hydrogen bromide ($\text{HBr}(\text{l})$, 80.91g/mol) react together to give aqueous barium bromide ($\text{BaBr}_2(\text{aq})$) and liquid water ($\text{H}_2\text{O}(\text{l})$). Which species would be the limiting reagent of this reaction?

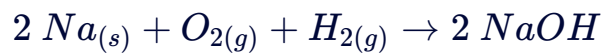
Example: How Many Grams of the Excess Reagent Will Remain?

The reaction between P_4 and Br_2 is very exothermic and produces PBr_5 as the only product. If 7.0 g of P_4 react with 12.0 g of Br_2 how many grams of the excess reagent will remain?

- a) 0.4 g
- b) 4.7 g
- c) 6.1g
- d)11.1 g

Practice: What is the Limiting Reagent

If 12 g of sodium is mixed with 0.64 g of H_2 and 17 g of O_2 which reagent will be limiting in the reaction shown below?



A) Na

☐

B) O_2

☐

C) H_2

☐

D) None, they will all be completely consumed

☐

4.5 Percent Yield

4.5.1

Percent Yield

So far we have been assuming that the reaction proceeds 100% to products. In reality, this is rarely the case!

When we are doing experiments in the lab we often lose some of our product by spilling it or residues of it get left on glassware and spatulas.



So far we have been calculating the **theoretical yield**.

Theoretical Yield - the maximum amount of product produced based on the quantity of the limiting reagent (e.g. the amount of product produced in a calculation).

Actual Yield - the actual amount of product produced in the reaction (e.g. the amount of product obtained in a laboratory experiment).

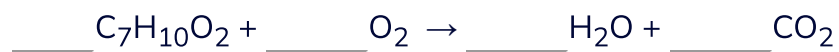
$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

Test Your Understanding

- 1) In a lab, you measure your product and record 5.222g. This is the _____ yield.
- 2) If everything went 100% perfectly according to the reaction we would get the _____ yield
- 3) If I determined the limiting reagent, and then calculated the number of moles of product and the mass of product based on this, I've just calculated the _____ yield

Example: What is the Percent Yield

When 49.00 g of a hydrocarbon fuel with formula $\text{C}_7\text{H}_{10}\text{O}_2$ is reacted with excess oxygen, a total of 21.56 g of water is collected. What was the percent yield of the reaction?



Practice: Calculate Moles Given Percent Yield

If the yield for the following reaction is 45%, how many moles of KClO_3 are needed to produce 1 mol of O_2 ?



A) 0.4 moles

☐

B) 1.1 moles

☐

C) 1.5 moles

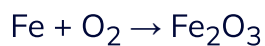
☐

D) 4.6 moles

☐

Practice: Limiting Reagent & Percent Yield

Iron oxidizes when exposed to oxygen to form iron oxide according to the following equation:



352g of pure iron is exposed to 12.0 mols of O_2 , and after a period of time 46.7 g of iron oxide (rust) is collected.

Part 1

What is the limiting reagent in this reaction?

A) Fe

☐

B) O_2

☐

C) Fe_2O_3

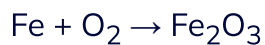
☐

D) There is no limiting reagent

☐

Practice: Limiting Reagent & Percent Yield

Iron oxidizes when exposed to oxygen to form iron oxide according to the following equation:



352g of pure iron is exposed to 12.0 mols of O_2 , and after a period of time 46.7 g of iron oxide (rust) is collected.

Part 2

What is the percent yield?

A) 2.2%

☐

B) 8.8%

☐

C) 9.4%

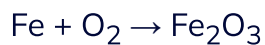
☐

D) 33.1 %

☐

Practice: Limiting Reagent & Percent Yield

Iron oxidizes when exposed to oxygen to form iron oxide according to the following equation:



352g of pure iron is exposed to 12.0 mols of O_2 , and after a period of time 46.7 g of iron oxide (rust) is collected.

Part 3

How many Fe atoms are present in the rust produced?

A) 0.89×10^{17} atoms

☐

B) 1.77×10^{23} atoms

☐

C) 3.56×10^{23} atoms

☐

D) 4.33×10^{23} atoms

☐

4.6 Titrations

4.6.1

Titration Problem Introduction

Titration involves reacting an acid and a base together to determine the concentration of the unknown (can be the acid or base).

WIZE TIP

On a test, you could be expected to know what the equivalence point of a titration is.

Equivalence point: is the point in a titration when enough titrant has been added into the reaction to completely react with the analyte.

In other words, the **acid and base completely react together** so that there is **no more OH⁻ or H⁺** as it would have all reacted.

Example:

We may be given the concentration and volume of the base and only the concentration of the acid and asked what volume of acid was needed to reach the equivalence point?

We will need to use the equation: $n=cv$ (n= moles, c=concentration, v=volume in L)

We could solve for the moles of the base ($n=cv$) and then since we are at equivalence point, we can figure out the moles of the acid.

i WIZE TIP

At this step, it is very important to write out a balanced chemical reaction with the acid and base reacting.

We must **pay attention to the coefficients** in the reaction as they are important for determining moles.

Then we could solve for the volume of the acid (use $n=cv$, rearrange for v: $v=n/c$)

Example: Titration Problem

A 100 mL solution of HBr was titrated with 0.20 M $\text{Ca}(\text{OH})_2$. It was found that 38 mL of the $\text{Ca}(\text{OH})_2$ solution was required to reach the equivalence point. What was the concentration of the HBr solution?

1) I'll start by writing out the variables that I know:

HBr:

$\text{Ca}(\text{OH})_2$

2) We can find the moles of $\text{Ca}(\text{OH})_2$ using $n=cv$

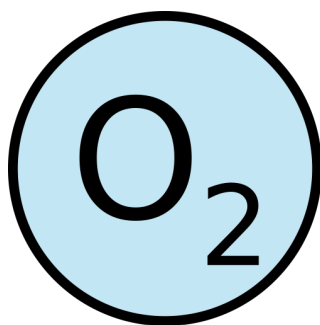
3) Now that we have moles of $\text{Ca}(\text{OH})_2$, we can figure out moles of HBr based on the balanced chemical equation

4) Now we can solve for the concentration of HBr

4.7 Classifying Reaction Types

4.7.1

Combustion Reactions



Combustion reaction: a reaction where a compound reacts with oxygen to produce the oxides of the elements that make up the compound

Complete vs Incomplete Combustion:

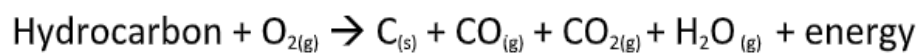
Complete Combustion: when an **excess of oxygen** reacts with a hydrocarbon to produce $\text{CO}_2 + \text{H}_2\text{O}$. This reaction releases energy.



Example:

Balance the following equation: $\underline{\hspace{1cm}} \text{C}_3\text{H}_8(\text{g}) + \underline{\hspace{1cm}} \text{O}_2(\text{g}) \rightarrow \underline{\hspace{1cm}} \text{CO}_2(\text{g}) + \underline{\hspace{1cm}} \text{H}_2\text{O}(\text{g})$

Incomplete Combustion: when there is **insufficient oxygen** present and the oxygen reacts with a hydrocarbon to form $\text{CO}_2 + \text{CO} + \text{H}_2\text{O} + \text{C}$. Here, the elements in the fuel don't combine with oxygen to the greatest extent possible.



Note: you can't balance incomplete combustion reactions :)

Classifying Reactions

The following are the options for classifying reactions:

- Combination
- Decomposition
- Single displacement/replacement
- Double displacement/replacement aka metathesis
- Combustion



ans: _____



ans: _____



ans: _____



ans: _____

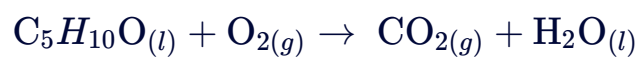


ans: _____

Example: Balancing

Balance the following reaction:

What type of reaction is this? _____



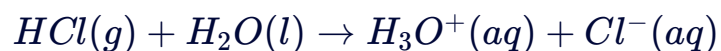
4.8 Acids and Bases

4.8.1

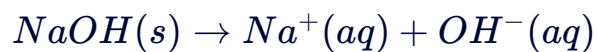
Definitions of Acids and Bases

Arrhenius definition: compounds identified as producing H^+ or OH^- ions:

- **Acids** produce _____ aka _____ :



- **Bases** produce _____ aka _____ ions:

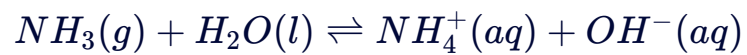


****Bronsted - Lowry definition:** compounds identified as H^+ acceptors or donors

- **Acids** _____ H^+ ions



- **Bases** _____ H^+ ions

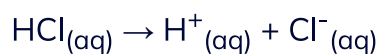


Acids

Strong Acids

Strong acids **dissociate completely**.

Example:



WIZE TIP

The following are strong acids that you should memorize!

- HCl (hydrochloric acid)
- H_2SO_4 (sulfuric acid)
- HNO_3 (nitric acid)
- HClO_4 (perchloric acid)
- HBr (hydrobromic acid)
- HI (hydroiodic acid)

Weak Acids

Weak acids **dissociate incompletely**.

Bases

Strong Bases

Strong bases **dissociate completely**.

Example:



WIZE TIP

The following are strong bases that you should memorize!

- All Group 1 Hydroxides (ex. NaOH, KOH, RbOH, CsOH)
- 3 Group 2 Hydroxides (Ca(OH)₂, Sr(OH)₂, Ba(OH)₂)
- All Group 1 Oxides (ex. Li₂O, Na₂O, K₂O)
- Metal amides (ex. M-NH₂ where M is a metal)

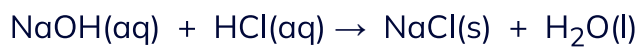
Weak Bases

Weak bases **dissociate incompletely**.

Neutralization Reactions

Neutralization reaction: a reaction where an acid and a base react together to produce a salt and water

Example:



4.9 Conjugate Acid-Base Pairs

4.9.1

Identifying Conjugate Acid and Base Pairs

Two substances that **differ from each other only by one proton (H⁺)** are referred to as a **conjugate acid-base pair**.

Identify the acid, base, conjugate acid, and conjugate base in the following reactions:



Write the K_a or K_b expression for this reaction:

- If we were to increase K_a , would that lead to more/less dissociation into ions? _____
- Would this mean we have a stronger/weaker acid? _____



Write the K_a or K_b expression for this reaction:

- If we were to increase K_b , would that lead to more/less OH^- ions? _____
- Would we have a stronger/weaker base? _____

Practice: Conjugate Acid-Base Pairs

In the reaction $\text{HSO}_4^-(\text{aq}) + \text{OH}^-(\text{aq}) \rightleftharpoons \text{SO}_4^{2-}(\text{aq}) + \text{H}_2\text{O}(\text{l})$, the conjugate acid-base pairs are:

HSO_4^- and OH^- ; SO_4^{2-} and H_3O^+

☐

HSO_4^- and SO_4^{2-} ; H_2O and OH^-

☐

HSO_4^- and OH^- ; SO_4^{2-} and H_2O .

☐

HSO_4^- and H_2O ; OH and SO_4^{2-}

☐

HSO_4^- and H_3O^+ ; SO_4^{2-} and OH^-

☐

4.10 Oxidation States

4.10.1

Oxidation States

Oxidation state: A way to describe the degree of oxidation of a chemical

WIZE TIP

Tips to Determine Oxidation State:

1. Free elements: oxidation state = 0

Examples: $\text{Cl}_2(\text{g})$ has an oxidation state of 0, $\text{Na}(\text{s})$ has an oxidation state of 0

2. Monoatomic ions: oxidation state = ionic charge

Examples: $\text{Cl}^-(\text{aq})$ has an oxidation state of -1, $\text{Na}^+(\text{aq})$ has an oxidation state of +1.

3. Hydrogen: oxidation state = +1

α. Exception: hydrogen in hydrides (e.g., NaH) – oxidation state = -1

4. Oxygen: oxidation state = -2

α. Exception: oxygen in peroxides (e.g., H_2O_2) – oxidation state = -1

5. For neutral molecules, oxidation numbers add up to zero.

For polyatomic ions, oxidation numbers add up to the ion charge.

Examples: $\text{Cl}_2(\text{g})$ has an oxidation state of 0, NaCl oxidation numbers add up to 0, SO_4^{2-} oxidation numbers add up to 2-.

Oxidation States - Additional Information

Other Low Yield Tips to Help Determine Oxidation States

- Alkali metals: oxidation state= +1
- Alkaline earth metals: oxidation state= +2
- Nitrogen Group: oxidation state= -3
- Oxygen family: oxidation state= -2
- Halogens: oxidation state= -1

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu			
Actinides	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr			

Note on Transition Metals

Transition metals can have different charges (ex. Pb^{2+} and Pb^{4+})
As a result, transition metals can also have different oxidation states!

Example:

What is the oxidation state of Cu in CuCl? _____

What is the oxidation state of copper in $\text{Cu}(\text{NO}_3)_2$? _____

When naming compounds involving transition metals, the **oxidation state of the transition metal is written in Roman numerals and put in brackets:**

- copper(I) chloride: CuCl
- copper(II) nitrate: $\text{Cu}(\text{NO}_3)_2$

Example: Oxidation State Practice

What is the oxidation state of:

S in S_2O_3 : _____

O in S_2O_3 : _____

What about Pb in Pb_2O_3 ?

This one is different because this compound is a mixed oxidation state compound, so although Pb should have 2+ or 4+, here, we get an oxidation state of _____ and that is ok because both Pb^{2+} and Pb^{4+} are present in the compound.

4.10.4

What is the oxidation state of Br in BrO_3^- ?

+4

☐

+5

☐

+6

☐

-4

☐

-5

☐

-6

☐

4.10.5

What is the oxidation state for the following: (enter answer as -1 or +2 for example)

a) P in P_2O_5 :

b) Cr in $\text{Na}_2\text{Cr}_2\text{O}_7$:

c) Al in $\text{Al}_2(\text{SO}_4)_3$:

d) Na in Na_2O_2 :

e) O in Na_2O_2 :

f) N in NH_3 :

4.11 Reduction-Oxidation (Redox) Reactions

4.11.1

Redox Reactions

OIL RIG! Oxidation Is Loss of electrons, Reduction Is Gain of electrons

OR

LEO GER! Loss of Electrons is Oxidation, Gain of Electrons is Reduction

OXIDATION (lose e^-)

REDUCTION (gain e^-)

Reducing agent reduces others and is itself oxidized.

Oxidizing agent oxidizes others and is itself reduced.

Oxidation number increases.

Oxidation number is reduced.

Keep in mind that in organic chemistry the definitions of oxidation and reduction are different:

- For **reduction**, we would look for **more bonds to H** and **less bonds to O**
- For **oxidation**, we would look for **less bonds to H** and **more bonds to O!**

How can you recognize if a redox reaction is occurring?

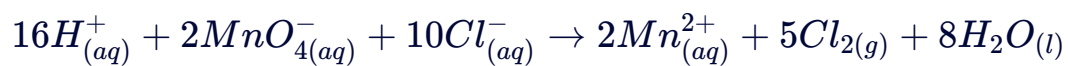
1. Can **check oxidation #s of each element** to see if they **change from reactants to products** (most important for now!)

2. Familiarize yourself with **common oxidizing and reducing agents** (you'll learn more about this in orgo!)

- **Oxidizing agents:** O_2 , halogens, HNO_3 , $\text{Cr}_2\text{O}_7^{2-}$, MnO_4^-
- **Reducing agents:** H_2 , metals, C

Practice: Oxidation vs Reduction

Consider the following balanced redox equation:



Which species is being oxidized?

☐ H^{+}

☐ MnO_4^{-}

☐ Cl^{-}

☐ Mn^{2+}

☐ Cl_2

4.12 Balancing Redox Reactions [half-reaction method]

4.12.1

Balancing Redox Reactions: Half-Reaction Method

The "half-reaction method" of balancing redox reactions is the most commonly used method :)

Balance the following redox reaction in basic conditions and assign oxidation numbers to all species.



WIZE TIP

Balancing redox equations:

1. **Check the oxidation numbers** of each species by inspection (**change in oxidation # means this is a redox reaction!**)
2. **Write** the reduction and oxidation **half-reactions**.
3. **Balance each** half-reaction: start with **atoms that are neither O nor H**, balance oxygens with H_2O , hydrogens with H^+ , and remaining charges with e^- .
4. **Multiply** the reduction and oxidation half-reactions as needed so that each reaction exchanges the **same number of e^-** .
5. **Add** the oxidation and reduction half reactions together and simplify.
6. **Additional step in basic solution only:** there can't be any H^+ ions, so **neutralize the H^+ by adding the same # of OH^- to each side**.

! WATCH OUT!

Always solve in acidic conditions first!

Then, if the question is asking for basic conditions do step #6!

1. Check the oxidation numbers of each species by inspection (change in oxidation # means this is a redox reaction!)

Here we are told it's a redox reaction already, but we can still check oxidation #s to see what is oxidized and what is reduced:



- Ag goes from having an oxidation # of _____ to _____. Therefore Ag is _____.
- Zn goes from having an oxidation # of _____ to _____. Therefore Zn is _____.

2. Write the reduction and oxidation half-reactions.

Oxidation Half Reaction:

Reduction Half Reaction:

3. Balance each half-reaction: start with atoms that are neither O nor H, balance oxygens with H_2O , hydrogens with H^+ , and remaining charges with e^- .

a) Balance everything that is not O and H:

ox:

red:

ox: $2\text{Ag(s)} \rightarrow \text{Ag}_2\text{O(aq)}$

red: $\text{Zn}^{2+}(\text{aq}) \rightarrow \text{Zn(s)}$

b) Balance O with H_2O :

ox:

red:

c) Balance H with H^+

ox:

red:

d) Balance charges with electrons!

ox:

red:

ox:

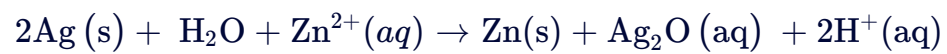


red:



4. Multiply the reduction and oxidation half-reactions as needed so that each reaction exchanges the same number of e⁻.

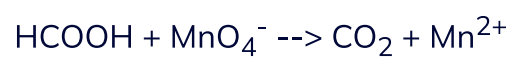
5. Add the oxidation and reduction half reactions together and simplify.



6. Additional step in basic solution only: there can't be any H^+ ions, so neutralize the H^+ by adding the same # of OH^- to each side.

Practice: Balancing Redox Reactions

Balance the following redox reaction in an acidic solution.



4.13 Precipitation Reactions

4.13.1

Solubility Chart

The following table tells us if certain combinations of ions are soluble or insoluble!



WIZE CONCEPT

If a combination of ions is **soluble**, it means that **no precipitate will form**.

If a combination of ions is **insoluble**, it means a **precipitate will form**!

You can expect to see a question on your exam that asks you if a precipitate will form or not!

Soluble Ionic Compounds	contain these ions	exceptions
	NH_4^+ group I cations: Li^+ Na^+ K^+ Rb^+ Cs^+	none
	Cl^- Br^- I^-	compounds with Ag^+ , Hg_2^{2+} , and Pb^{2+}
	F^-	compounds with group 2 metal cations, Pb^{2+} and Fe^{3+}
	$\text{C}_2\text{H}_3\text{O}_2^-$ HCO_3^- NO_3^- ClO_3^-	none
	SO_4^{2-}	compounds with Ag^+ , Ba^{2+} , Ca^{2+} , Hg_2^{2+} , Pb^{2+} and Sr^{2+}
Insoluble Ionic Compounds	contain these ions	exceptions
	CO_3^{2-} CrO_4^{2-} PO_4^{3-} S^{2-}	compounds with group 1 cations and NH_4^+
	OH^-	compounds with group 1 cations and Ba^{2+}

Photo by OpenStax / CC BY

Decide whether the following salts would form a precipate or not:

1) NaCl

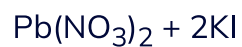
2) AgCl

3) AgI

4) NaNO_3

Example: Does a Precipitate Form?

In the following example, do you think a precipitate is formed? If yes, what is the formula of the precipitate?



Write the complete chemical reaction:



i WIZE TIP

In a chemical reaction, **(aq)** means that the substance is dissolved in aqueous solution (aka it is **soluble**),
whereas **(s)** means that the substance is in solid form and is **insoluble** in solution (aka **forms a precipitate**)

This reaction can be referred to as a **precipitation reaction**.

4.14 Net Ionic Equations

4.14.1

Three Ways to Write Out Chemical Reactions

There are 3 different ways to write out a chemical reaction.

1) Molecular Reaction

This is the way we are most familiar with!



2) Complete Ionic Equation

To write the complete ionic equation, we need to look at the molecular reaction and identify which of the molecules can dissociate into ions. Then write out the equation with all of the ions.

WIZE TIP

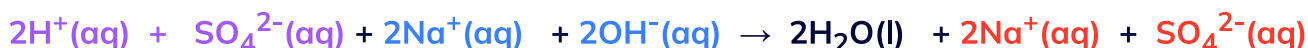
Things that won't dissociate include: solids, liquids, gases, weak acids, weak bases, and combinations of ions that we said are insoluble (from the solubility chart)

Things that will dissociate include: strong acids, strong bases, and combinations of ions that we said are soluble (from the solubility chart)

Compound in the molecular reaction:	$\text{H}_2\text{SO}_4(\text{aq})$	$2\text{NaOH}(\text{aq})$	$2\text{H}_2\text{O}(\text{l})$	$\text{Na}_2\text{SO}_4(\text{aq})$
Can this compound fully dissociate? If so, why?				
If the compound can fully dissociate, write out the balanced reaction to show this:				

[Solution available online](#)

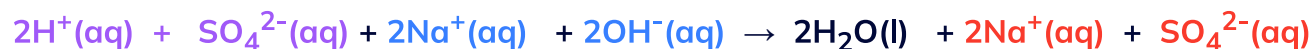
Next, since we figured out all the molecules that dissociate and balanced the equations, we just put all the ions into the equation so they replace the aqueous molecules:



3) Net Ionic Equation:

For this equation, we just take the complete ionic equation and cross anything out that appears in equal quantities on both sides so that we are left with the net equation.

Complete Ionic Equation:



and this can be further simplified to get the net ionic equation:

WIZE TIP

When we talk about ionic equations, you may see the term spectator ions used. **Spectator ions** are the ions that remain unchanged on both sides of the reaction. In this case, the spectator ions are SO_4^{2-} and Na^+ .